CPSC 368 Project - Group 4 - Phase 3

For each dataset used in your project, provide:

a. **The name/description of the dataset**
b. **The source/URL where we can find it**
c. **A brief (1-2 sentence) description of what value this dataset provides to your research question.**

1. Youtube Data API V3

- **Link:** https://developers.google.com/youtube/v3
- **Description:** API for accessing youtube data which can provide us information for channel-level statistics and video-level statistics and comment threads. We've made 3 tables from this API: *youtube_channels, youtube_videos, youtube_comments*
- **Revelance:** This dataset provides metrics such as view counts, likes, and comments for two political YouTube channels and videos (Ben Shapiro and The Young Turks. These channels were selected as these are the 2 biggest non-news outlets channels that lean Republican and Democrat respectively and with their data, we can analyze engagement trends and sentiment.

2. American National Election Studies Data

- **Link:** https://electionstudies.org/data-center/
- **Description:** Survey conducted before/after each presidential election detailing questions on political attitudes, media consumption, demographics, video behaviour and other questions indicative of political sentiment
- **Relevance:** This dataset allows us to compare self-reported ideology and media consumption with patterns of YouTube engagement and election outcomes. By identifying individuals who did and didn't report consuming political content on social media, we can determine whether voting patterns differ between the two groups and compare that to the trends observed on Youtube and election outcomes.

3. MIT Election Data + Science Lab

- **Link:** https://electionlab.mit.edu/data
- **Description:** Cleaned standardized election results for U.S federal elections including presidential, senate and house results for various geographic levels
- **Revelance:** This dataset provides election outcomes that allow us to compare political engagement on YouTube with real-world voting results.

<u>Describe the data cleaning steps you performed in Step 1 from the Workflow section. If you did not perform any data cleaning, briefly explain why.</u>

1. Youtube Data API V3

The YouTube data was collected directly from the YouTube Data API v3 and required minimal cleaning due to the clean data gathered from the API. Instead, a lot of the difficulty came from getting the information we needed from the API itself, but afterwards data came naturally.

The following steps were performed:

a. Videos were filtered to a shared window of January 2018 to December 2025, chosen because The Young Turks channel's uploads playlist only contained data from late 2017 onward. This ensured a consistent time range across both channels for valid comparison.
b. Found the channel statistics and gathered video stats on all the videos posted by these channels. From there, since there was over 15000 videos, we selected the top 5 most viewed for each month (96 months total), leaving us with 960 videos that we selected into our dataset.
c. Out of the 5 videos for each month per channel, we chose the top 3 most viewed and selected the top 3 most relevant comments. By most relevant, this is up to Youtube algorithms, but this was supposed to leave us with 2880 comments, but we had 1240 comments as these channels had very little following in the beginning

There were some transformations that were applied programmatically within our code, as working with the YouTube data was a bit finicky, but for the most part we left the data bare bones so that we could perform our transformations in SQL.

Additionally, in order to do comment sentiment analysis, we used the popular Python package **vaderSentiment** that is commonly used for social media comments.

American National Election Studies Data

The ANES data was presented with 800 page code books. To make the data easier to work with without consistently referring to the manuals, variable names and values were renamed to human readable values. Within each variable, the data used various values to represent different reasons for data that could not be collected. With the exception of those who answered “don’t know” or “uncertain”, other reasons for missing data were converted to NaN as they were not the focus of this report and this would make the data easier to use with Python’s statistical packages.

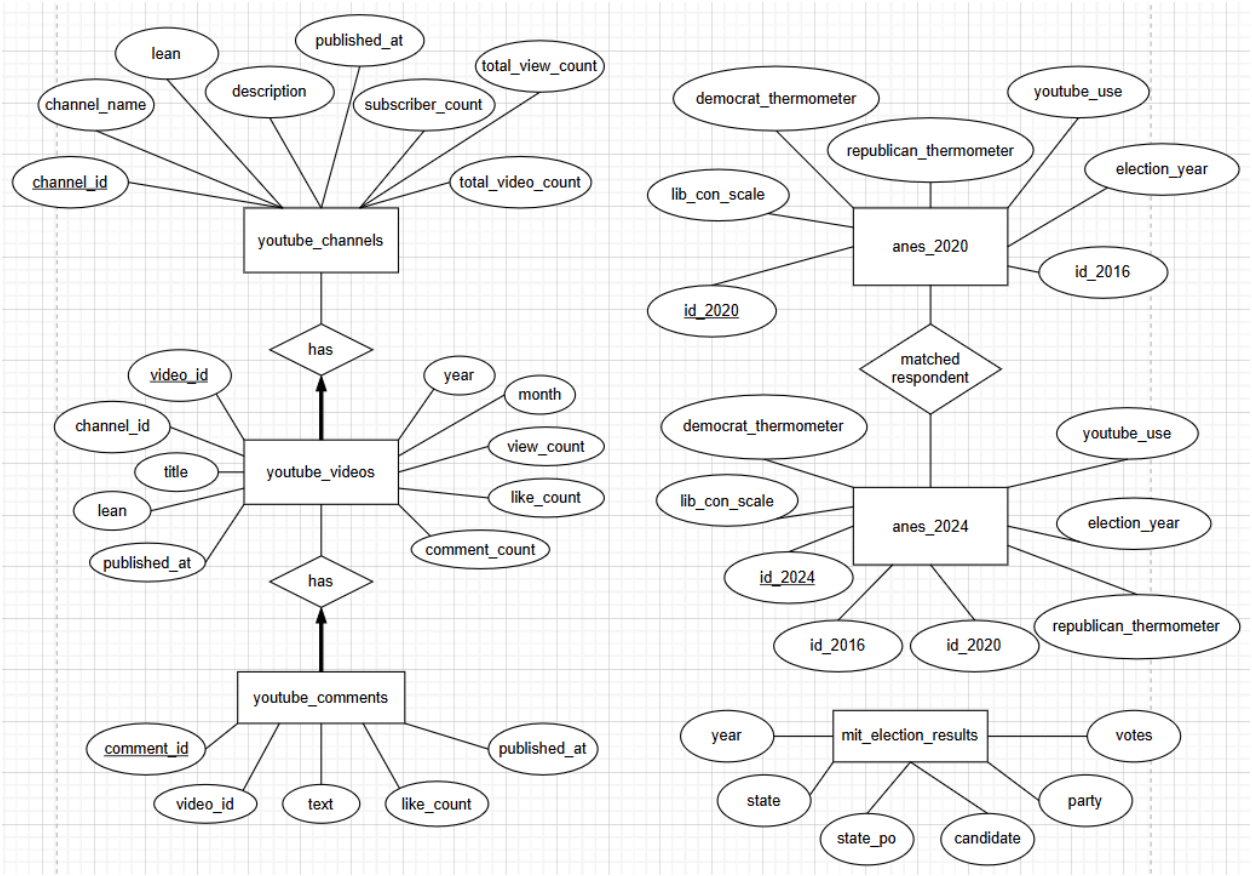
- Filter to get two tables ‘anes_2020’ and ‘anes_2024’ and standardize ‘election_year’ column

- Renaming variables for better readability for all columns at both tables (e.g: ‘V240001’ to ‘id_2024, ‘V242577e’ to ‘youtube_use’)
- Convert numeric codes (due to missing/invalid responses) to NaN for ‘id_2016’, ‘id_2020’, ‘id_2024’, ‘democrat_thermometer’, ‘republican_thermometer’, and ‘youtube_use’
- Convert numeric response codes into readable category labels for ‘lib_con_scale’ and ‘youtube_use’
- Randomly sample 500 observations for each table

MIT Election Data + Science Lab

- Filter rows to get Democrat and Republican candidates
- Drop rows where ‘candidate’ column was missing
- Select ‘year’, ‘state’, ‘state_po’, ‘candidate’, ‘party’ (renamed from ‘party_simplified’), and ‘votes’

ER diagram from Step 2 of the Workflow section.



State your overarching research question for your project. If you wish, you can reuse what you had for prior phases without rewording it.

This study seeks to explore the relationship between political media content, election outcomes, and overall voters' behavior. The three guiding research questions are:

- How does average viewership and engagement of conservative and liberal YouTube channels change over time?
- Do YouTube users lean more towards a particular political party than non-users, and does this relationship change across election cycles?
- Does YouTube comment sentiment reflect voter preferences or amplify extreme views?

<u>Research Question 1) How does average viewership and engagement of conservative and liberal YouTube channels change over time?</u>

SQL:

```
SELECT lean, year,
ROUND(AVG(view_count), 2) AS avg_views,
ROUND(AVG(like_count), 2) AS avg_likes,
ROUND(AVG(comment_count), 2) AS avg_comments,
ROUND(AVG(like_count) / AVG(view_count) * 100, 2) AS like_rate_percent,
ROUND(AVG(comment_count) / AVG(view_count) * 100, 2) AS comment_rate_percent
FROM youtube_videos
GROUP BY lean, year
ORDER BY year, lean
```

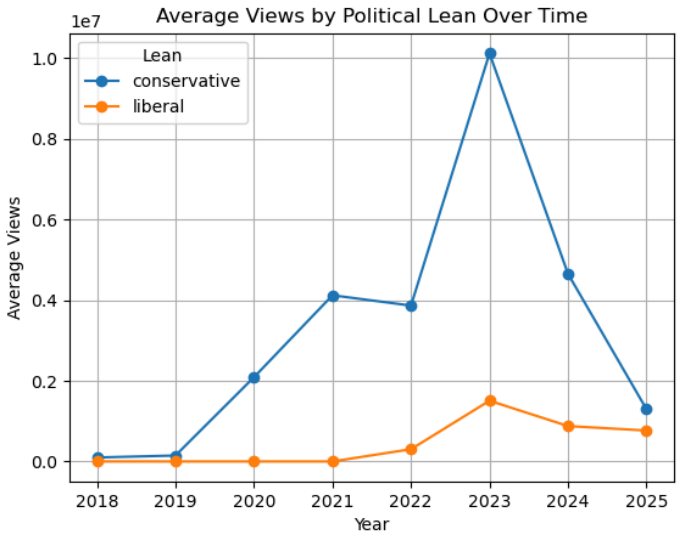
Python File & Line Number:

https://github.com/calvingdu/cpsc368/blob/master/sql/research_questions.ipynb

Query Explanation:

This query looks at average metrics such as views, like and comments and shows how these have changed over time. We chose this query as it let us see the time trends of engagement on the channels.

Table/Visual:



We focused on views for the visual and can see that leading up to the 2024 election (Trump/Conservative/Republican win), that there was a huge spike in Conservative channel views before the election. As expected, there's a big drop as the election wraps up.

Research Question 2) Do YouTube users lean more towards a particular political party than non-users, and does this relationship change across election cycles?

SQL:

```
SELECT
a.election_year,
a.youtube_use,
a.lib_con_scale,
COUNT(*) AS respondent_count,
ROUND(AVG(a.democrat_thermometer), 2) AS avg_dem_thermometer,
ROUND(AVG(a.republican_thermometer), 2) AS avg_rep_thermometer,
ROUND(AVG(a.democrat_thermometer) - AVG(a.republican_thermometer), 2) AS
partisan_lean
FROM anes_2020 a
WHERE a.youtube_use IS NOT NULL
AND a.lib_con_scale NOT IN ('DK', 'Refused')
AND a.lib_con_scale IS NOT NULL
GROUP BY a.election_year, a.youtube_use, a.lib_con_scale
UNION ALL
SELECT
a.election_year,
a.youtube_use,
a.lib_con_scale,
COUNT(*),
ROUND(AVG(a.democrat_thermometer), 2),
ROUND(AVG(a.republican_thermometer), 2),
ROUND(AVG(a.democrat_thermometer) - AVG(a.republican_thermometer), 2)
FROM anes_2024 a
WHERE a.youtube_use IS NOT NULL
AND a.lib_con_scale NOT IN ('DK', 'Refused')
AND a.lib_con_scale IS NOT NULL
GROUP BY a.election_year, a.youtube_use, a.lib_con_scale

ORDER BY election_year, youtube_use, lib_con_scale
```

Python File & Line Number:

https://github.com/calvingdu/cpsc368/blob/master/sql/research_questions_2.ipynb

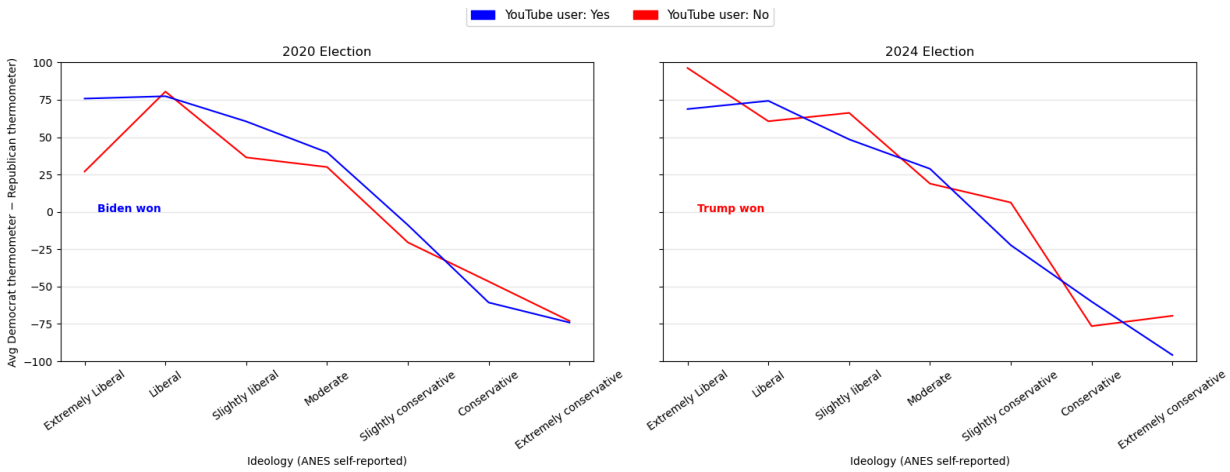
Query Explanation:

For each election year and ideology group, we compare the political lean of YouTube users vs non-YouTube users (taken from the survey) to see if they systematically differ in their political preferences. The plot basically compares that when we quantify the ideological strength of (democrat - republican) per survey respondent, how does it look when we compare Youtube users vs Nonyoutube users. We can then see if this changes per election year.

The WHERE clause filters out respondents who answered "DK" (don't know) or "Refused" on the ideology scale, and drops anyone who didn't respond to their youtube use. This keeps only respondents that we can compare.

The GROUP BY creates one row per unique combination of election year, YouTube use (Yes/No), and a label for ideology.

Table/Visual:



In this visual we can see that the trend is generally consistent, but this is expected as it should be a downward slope. The most interesting thing here is that Youtube users are consistently more moderate than non-users across every group (especially in 2020). In 2024 the lines are closer together and cross multiple times showing that Youtube vs Non-Youtuber differences diminish. This could indicate that Youtube has become more politically Conservative/Republican in 2024, suggesting that there is more Red-wing influence. This lines up with what we know about politics, as many suggest that Republican views have gotten more vocal in recent times.

<u>Research Question 3) Does YouTube comment sentiment reflect voter preferences or amplify extreme views?</u>

SQL:

```
SELECT ch.channel_name, v.lean, v.year,
ROUND(AVG(c.sentiment_score), 3) AS avg_sentiment,
COUNT(*) AS comment_count
FROM youtube_comments c
JOIN youtube_videos v ON c.video_id = v.video_id
JOIN youtube_channels ch ON v.channel_id = ch.channel_id
GROUP BY ch.channel_name, v.lean, v.year
ORDER BY v.year, v.lean, ch.channel_name
```

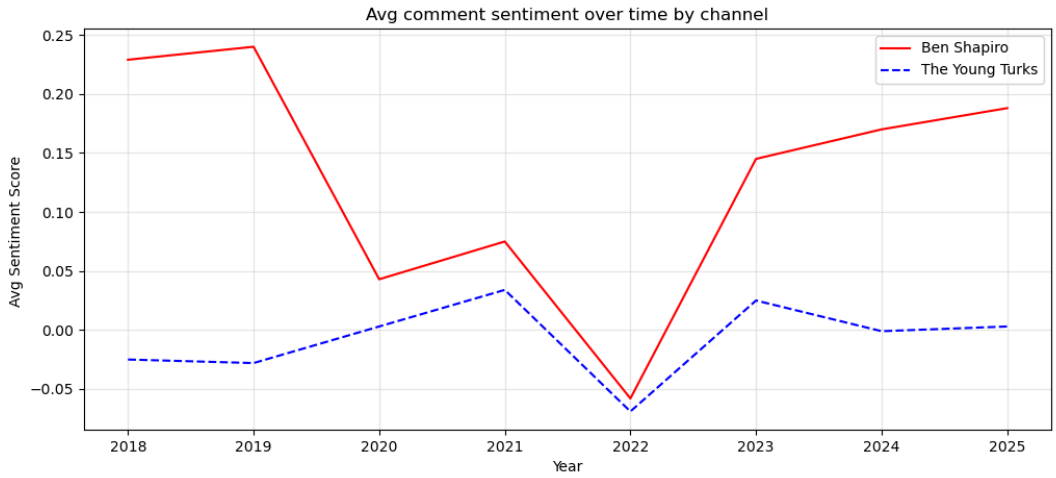
Python File & Line Number:

https://github.com/calvingdu/cpsc368/blob/master/sql/research_questions_3.ipynb

Query Explanation:

This query takes for each channel, what are the comments sentiment over time?

Table/Visual:



Something quite interesting is that the lowest sentiment across both channels is in 2022. 2022 was a midterm election year with high political tensions, so seeing both views as extremely negative makes sense. What's most interesting here is that in 2020, we see a huge drop in sentiment in Ben Shapiro's comments and sentiment increased in The Young Turks which makes sense due to the Trump loss. Then we can see in 2023 to 2024, Ben Shapiro's become increasingly more positive and The Young Turk's negative. What we can conclude here is that sentiment does partially reflect how the public sees political events. But additionally, we can see

that Ben Shapiro’s comments are consistently more positive no matter the event (2020 Republican loss and 2024 Republican Win) indicating that while they are affected by events, it seems that channel culture is what is affecting the sentiment more. This may indicate that there is some amplification of views here, as each comment’s section's emotional baseline persists despite the political event outcomes.

AI Tool Use Declaration: Refer to the Project Overview document for expectations.

Used AI to help understand API functionality for Youtube Data